Oséias Dias de Farias

Machine Learning & AI Engineer | MLOps

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Summary

Machine Learning and AI engineer with a background in electrical engineering and industrial maintenance. I build AI systems that keep physical systems running: they detect when something fails and help fix it, in production. Today I build fault-tolerant control software in a research project funded by the U.S. Air Force Office of Scientific Research (AFOSR) at UFABC, where I am an M.Sc. candidate. Before that, I shipped voice AI and multi-tenant integrations for an AI agent platform on Java (Quarkus), Vue.js and AWS serverless. Author of Synapsys, an open-source control and ML framework on PyPI with 501 tests and 100% coverage.

Experience

AI Engineer & Researcher **Apr 2025 – Present**

Federal University of ABC (UFABC) · AFOSR-funded Artificial Anti-fragility project · Santo André, Brazil

- Built a JADE/FIPA multi-agent supervisor for a DC-DC buck converter: anomaly detection (Luenberger observer, adaptive CUSUM, windowed RMSE), online ARX identification and autonomous controller reconfiguration (LQR via DARE or H_∞ synthesis) with a bumpless hot swap. In software-in-the-loop tests, RMSE dropped **60.8%** (nominal) and **33.3%** (stochastic parameter stress), reconfiguring in **under 500 ms**. First author, CBA 2026 (accepted).
- Built the plant-side software: Simulink buck model with control, fault detection and isolation and an antifragility index in Python, coupled by software-in-the-loop (C S-Function, ZeroMQ) and streaming telemetry to the AI supervisor.
- Designed distributed Python components with ZeroMQ messaging, shared-memory ring buffers (posix_ipc, NumPy), Pydantic v2 data contracts and Pytest suites; wrote drivers for hardware-in-the-loop validation (FPGA/FPAA).

Backend & AI Developer **Dec 2025 – Jun 2026**

Verzel · Lunella AI agent platform · São Paulo, Brazil (Remote)

- Built a bring-your-own-key voice feature end to end (Vue.js, Quarkus, database migrations): customers store their own ElevenLabs API key, encrypted at rest, and preview their available voices.
- Integrated VAPI into a multi-tenant AI agent platform: per-workspace credentials, per-tenant tools, Twilio phone number provisioning and an IVR that transfers calls between specialist agents.
- Connected the Quarkus back end (GraphQL/Apollo) to AWS serverless services (Lambda, SQS, DynamoDB, API Gateway), deployed with Serverless Framework and Bitbucket CI.
- Worked on a 9+ microservice platform: LLM agents with TaskingAI, a reactive Java back end (Hibernate Panache, PostgreSQL with Flyway, Keycloak OIDC, Redis) and Python services with FastAPI, Docker and Pytest.

Electrical Engineering Intern **Sep 2023 – Mar 2024**

Grupo Piracanjuba · UHT maintenance · Bela Vista de Goiás, Brazil

- Led a data analysis project on historical downtime (Python, Pandas) to find the root causes of unplanned stops on three production lines and proposed low-cost jigs to standardize line changeovers.

Selected Projects

- **Synapsys** (open source, PyPI): Python framework that joins classical control and ML: PyTorch/Keras/JAX models as controllers, distributed multi-agent simulation (shared memory, ZeroMQ), MATLAB-style API, 3D physics simulators and MIL/SIL/HIL workflows. **501 tests, 100% coverage.** github.com/synapsys-lab/synapsys
- **Youfy** (MLOps lab): DVC-versioned audio features, artist-disjoint splits verified with property-based tests, deterministic training tracked in MLflow, a model registry with a seed-noise-aware promotion gate and a serving loader that rejects feature-spec mismatches; 109 tests in CI. github.com/Oseiasdfarias/youfy

- **sdr-agent**: LLM sales agent on the OpenAI Assistants API with tool calling to a CRM (Pipefy) and a calendar (Cal.com); serverless FastAPI back end, Redis sessions, React chat. GitHub repository
- **Otensor** (founder): multi-tenant IoT automation platform (ESP32, Raspberry Pi, MQTT, AWS SAM, Lambda) with an event-driven automation engine, validated end to end on real hardware. docs.otensor.com.br

Publications

- **Farias, O. D.**, Estrada Martínez, E., Lescano, J. C. O. S., Potts, A. S. *Supervisão Adaptativa de Conversor Buck por Sistema Multiagente com Ciclo Percepção-Raciocínio-Ação e Reconfiguração Autônoma de Controladores.* Brazilian Conference on Automation (CBA), 2026. Accepted.
- **Farias, O. D.**, Teixeira, R. B. *FCS-MPC com ARX Adaptativo via RLS para Controle P-Q Robusto a Variações Paramétricas em Inversores Trifásicos Conectados à Rede.* Brazilian Conference on Automation (CBA), 2026. Accepted.

Education

M.Sc. in Electrical and Information Engineering **2025 – 2027 (expected)**
Federal University of ABC (UFABC), Brazil
Verifiable SIL/HIL integration of control, fault diagnosis and AI supervision, within the AFOSR-funded Artificial Anti-fragility project.

M.Sc. in Applied Computing **2025 – 2027 (expected)**
Federal University of Pará (UFPA), Brazil

B.Sc. in Electrical Engineering **2017 – 2024**
Federal University of Pará (UFPA), Brazil
Thesis: physical prototype and 3D digital twin of an aeropendulum for an open virtual control lab.

Technical Skills

Languages: Python, Java, SQL, C/C++, MATLAB
AI & ML: PyTorch, TensorFlow, scikit-learn, LLM agents and tool calling (TaskingAI, OpenAI Assistants), voice AI (ElevenLabs, VAPI, Twilio), reinforcement learning (DQN), anomaly and fault detection
MLOps: MLflow (tracking, model registry), DVC, Docker, CI/CD (GitHub Actions, Bitbucket), Pytest, Hypothesis
Cloud & back end: AWS (Lambda, SQS, SNS, DynamoDB, API Gateway, IAM, S3), Serverless Framework, AWS SAM, Quarkus, FastAPI, GraphQL, PostgreSQL, Redis, Keycloak
Control & embedded: Simulink, SIL/HIL, LQR, H_∞, Kalman filters, CUSUM, ZeroMQ, ESP32, Raspberry Pi, MQTT

Teaching & Writing

Udemy instructor (2 Python courses, 2,100+ students) · YouTube channel @otensor (1.7k subscribers) · 16-article series on MLOps and AI engineering on Medium

Languages

Portuguese (native) · English (technical reading)